That difficulty admits two responses, and they are routinely conflated. The first is robustness: train across a randomised family so that a single fixed model is adequate everywhere. It costs nothing at run time and gives up the difference between the family and the individual. The second is adaptation: infer which member of the family is present from the interaction history and condition the model or the policy on that estimate. It promises to recover that difference, at the cost of an estimator that must work from observable signals alone, since a deployed machine has no access to its own plant parameters.

The second response is only worth pursuing under conditions that are rarely checked before the work is done. Plant identity must be recoverable from the data the machine actually produces, which depends on how the machine is being driven while that data is collected. And the difference between knowing the plant and not knowing it must be large enough to matter once the estimate reaches a feedback controller. Neither condition is guaranteed, both are specific to the task, and both can be measured before an estimator is built. Establishing those measurements, and applying them to a hydraulic mowing arm is the problem this thesis addresses.

1.4 Contribution of This Thesis

The difficulty described in Section 1.3 is a modelling problem before it is a control problem. A hydraulic actuator can be simulated accurately, but the models that do so are too slow to sit inside a reinforcement-learning loop that requires millions of environment steps. The alternative usually taken is to discard the hydraulics and command joint positions directly, which removes the load and pressure dependence that makes the machine difficult to control. What is needed is a representation that is cheap enough to train against and faithful enough that the resulting policy is controlling something close to the real actuator.

Learned actuator models and latent-conditioned adaptation are both established techniques and Section 1.2 described the progress that makes them practical on machines of this class. What remains unclear is how such models should be identified so that they retain the command dependence a controller needs, whether they transfer between simulators without being retrained, and whether online adaptation adds anything once a policy has been trained against them. This thesis therefore treats representation and transfer as the primary questions, and treats online adaptation as the last of them rather than the first. The research statement of the thesis is as follows:

Comparison notation.

For readability, the three policy–actuator stacks are denoted by short labels throughout this thesis: *R* is PPO–ROM, *IF* is PPO–ID–FD, and *AIF* is PPO–(AAM)–ID–FD. R uses the command-conditioned reduced-order model; IF uses the frozen inverse/forward pair and AIF augments IF with online adaptation. The labels name the complete stacks, not just their actuator models.

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To determine how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation improves on the robustness those learned models already provide.